AZUSA PACIFIC UNIVERSITY

EFFECTS OF IMPLEMENTING TRX EXERCISES IN THE REHABILITATION OF A MALE FOLLOWING TOTAL HIP REPLACEMENT SURGERY

by

Angelina Browning

A capstone project submitted to the

School of Behavioral and Applied Sciences

in partial fulfillment of the requirements

for the degree Doctor of Physical Therapy

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DEDICATION

I would like to dedicate this to all my friends and family, especially my roommate, Lorena, who helped me through this process.

ACKNOWLEDGMENTS

I would like to thank the faculty at Azusa Pacific University, in particular Annette Karim and my mentor, Derrick Sueki, who guided me through this process. In addition, I would like to thank my sister for motivating me and for making me laugh through this process and my classmates for their support. Lastly, I would like to thank my CI, Nathan, and the patient who participated in this study.

ABSTRACT

**EFFECTS OF IMPLEMENTING
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OF A MALE FOLLOWING TOTAL HIP REPLACEMENT SURGERY**

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Doctor of Physical Therapy, 2018
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Background. Research has shown that osteoarthritis (OA) is the primary reason for Total Hip Replacement (THR) surgery. THR rehabilitation protocols vary depending on the facility and the surgeon, and limited research is available comparing these protocols to determine which protocols are the most beneficial.

Purpose. The purpose of this study was to evaluate a patient 6 weeks post-operative THR surgery and determine if Total Body Resistance Training (TRX) was more beneficial in improving hip strength, hip range of motion (ROM), balance, and gait speed, decreasing pain, and overall improving lower extremity (LE) function than usual care.

Case description. The patient was a 60-year-old male who developed OA of his left hip and underwent THR surgery to decrease pain and improve LE function.

Literature review. Current literature supports the use of closed-chain exercises in post-operative THR treatment and suggests that the most benefits are noted with the use of functional weight-bearing exercises.

Outcomes. In this case report, clinically significant improvements were noted in passive and active hip flexion and hip extension ranges of motion. Improvements were noted in active hip abduction range of motion, but not passively. Clinically significant strength improvements were noted in both gluteus maximus and medius muscles. Balance improved in the Single Limb Stance (SLS) test with eyes open but not closed. In the Sharpened Romberg Test, significant improvements were noted with eyes closed. In the Lower Extremity Functional Index (LEFI), the patient's score increased by nine points, a clinically significant improvement in function. No significant improvements were noted in pain using the Visual Analogue Scale (VAS).

Discussion. Research suggests that, following THR surgery using the posterior approach, patients demonstrate decreased hip strength due to compromised muscles during the procedure. Deficits in hip strength have negative effects on balance and can alter gait mechanics, thereby decreasing gait speed. From the literature review and this case report on THR rehabilitation, it is beneficial to begin treatment early on and incorporate a proper strength-training program.

Keywords: elderly; total hip replacement; total hip replacement protocol; TRX; closed-chain exercises, lower extremity function

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CHAPTER 1

INTRODUCTION

Osteoarthritis (OA) is a chronic condition where reactive changes occur in the bone and the cartilage between joints, commonly affecting the knees, hips, hands, and spine (Dagenais, Garbendian, & Wai, 2009). This degeneration is common in the elderly population and causes pain and stiffness and eventually can lead to disability. It has been predicted that hip OA will become a leading cause of disability globally by the year 2020 (Coulter, Scarvell, Neeman, & Smith, 2013). Symptomatic OA is the primary indication for a total hip replacement (THR; Bahl et al., 2018), and the demand is expected to increase (Di Monaco & Castiglioni, 2013).

The purpose of this study was to evaluate a patient who underwent THR surgery for symptomatic OA and to determine if Total Body Resistance Exercises (TRX), a form of suspension training based on functional exercises, are more effective in improving hip strength, hip range of motion (ROM), balance, and gait speed, decreasing pain, and overall improving lower extremity (LE) function compared to usual care.

Chapter 2 reviews literature available on rehabilitation post-THR surgery. One study observed deficits in gait mechanics after THR surgery and compared deficits prior to surgery as well as compared to healthy individuals. This study concluded that improvements were noted post-surgery, although deficits were still observed when compared to healthy individuals (Bahl et al., 2018). A systematic review assessed different THR treatment techniques and determined that, in the early phases of rehabilitation, benefits were noted with ergonomic cycling and with maximal strength training. In the late stages of rehabilitation, benefits were noted with exercises performed in weight bearing (Di Monaco & Castiglioni, 2013). A randomized controlled trial suggested that a physiotherapy-led functional program was more effective in improving gait speed compared to usual care (Monaghan et al., 2017). The effects of functional closed-chain exercises were assessed in the acute hospital setting. This study suggested that closed-chain functional exercises are safe and more effective than usual care in improving bed mobility, transfers, gait, and stair training—and, therefore, in decreasing hospital length of stays (Abbas & Daher, 2017). Currently, there is no research supporting the benefits of TRX training in improving clinical outcomes in the THR population.

Chapter 3 is a case report, which evaluated clinical outcomes for the rehabilitation of a patient who underwent THR surgery. The patient was a 60-year-old male who presented to physical therapy with hip pain and disability and with a diagnosis of hip OA. He underwent a THR and returned to therapy to regain function of his lower extremity. A TRX training program was selected as the intervention, and outcomes were compared to usual care. After 4 weeks of TRX training, the patient demonstrated improvements in hip ranges of motion and hip muscular strength, with significant improvements in single-limb balance, gait speed, and on the lower extremity functional index (LEFI).

Chapter 4 discusses the results of the case study and explains which results yielded significant findings. The discussion addresses the limitations attached to the study and reports the possibility for future studies.

CHAPTER 2

LITERATURE REVIEW

Approximately 4-9% of adults who are over the age of 45 suffer from symptomatic osteoarthritis (OA; Bahl et al., 2018). The prevalence of a total hip replacement in 2010 was 0.83% of the U.S. population, estimated as 2.5 million individuals (Kremers et al., 2015). Hip OA leads to pain and eventually disability; it is predicted that hip OA will become a leading cause of disability globally by 2020 (Coulter et al., 2013). Recent studies have noted that, prior to total hip replacement (THR) surgery, many patients with hip OA exhibit deficits in muscular strength in the affected limb when compared to their healthy limb. Deficits in strength and in postural stability continue to remain even 2 years post-THR surgery (Okoro, Lemmey, Maddison, & Andrew, 2012).

A literature review on rehabilitation post-THR revealed that post-operative exercise rehabilitation can potentially increase the rate of recovery and improve physical function (Gilbey et al., 2003). However, a specific and detailed protocol has not been agreed upon. Many protocols available are solely experienced based, and many are missing vital components, such as the duration of the intervention, the number of sessions, the specific equipment used, the time between each session, and intervention initiation time (Di Monaco & Castiglioni, 2013).

Suspension training is a type of functional training, primarily designed for rehabilitation and therapy due to its positive effects in strength, core muscle activation, and balance (Gaedtke, & Morat, 2015). Suspension training involves the use of anchored straps to suspend a specific body segment, creating an unstable environment. The individual works against his or her body weight, while adjusting the resistance by either moving closer to or further away from the anchor point (Smith, Snow, Fargo, Buchanan, & Dalleck, 2016). Although studies have examined the effects of suspension training, there is no research supporting the benefits of TRX suspension training in improving clinical outcomes in the THR population.

Purpose

The purpose of this literature review was to compare the clinical outcomes of a suspension-based protocol using Total Body Resistance Exercises (TRX) and traditional rehabilitation.

Methods: Evidence Acquisition

Data Sources and Search Strategies

In the months of February to March 2018, a comprehensive computer search was conducted on the rehabilitation of patients following a total hip replacement. The following electronic databases were used to conduct the search: MEDLINE, Pubmed, Cochrane, Physiotherapy Evidence Database (PEDro), and Google Scholar. The search terms used were *male*, *men*, *elderly*, *Total Hip Replacement*, *TRX*, *exercise*, *closed-chain exercise, Total Hip Replacement Protocol, pain, LEFI/ LEFS, AROM, PROM, strength, gait speed*, and *balance.* These terms were pertinent to the patient, intervention, comparison, and outcome (PICO) question. A single researcher conducted the search and screened the results for relevant and reliable articles (see Figure A1).

Study Selection

Articles pertinent to the rehabilitation of post-operative total hip replacements were included. To expand the search, articles that addressed rehabilitation post-arthroscopic surgery for femoral acetabular impingement (FAI) and labral tears were also included. Articles with relevant objective measures taken pre- and post-treatment with specific timelines were included. Those studies that were still in progress without results available were excluded from the search as well as research published prior to 10 years ago and articles that were not in English (see Figure A1).

Assessment of Methodologic Quality

Articles selected for this review were assigned a quality score. To evaluate systematic reviews, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) scale was used; Randomized Controlled Trials (RCTs) were scored using the Physiotherapy Evidence Database (PEDro) scale; observational studies were scored using the Case Reports (CARE) scale; and observational studies were evaluated with the Strengthening the Reporting of Observational studies in Epidemiology (STROBE) scale. Additionally, each article was assigned a Center for Evidence-Based Medicine (CEBM) Level of Evidence score (see Table A1). Higher-quality study designs are represented with lower CEBM scores.

Methods: Evidence Synthesis

Initially, 297 articles were screened, of which 24 articles were reviewed in full text. Of those 24 articles, only those that were peer reviewed were included in the study, and six duplicates were removed. A total of nine articles were included in the final review. The review includes four systematic reviews, two RCTs, one case report, and two observational studies.

Systematic Reviews

A systematic review compared 11 papers on nine RCTs to determine which type of exercise was effective following a THR. The systematic review concluded that maximal strength training and ergometer cycling resulted in favorable outcomes in the early post-operative phase (fewer than 8 weeks), and weight-bearing exercises yielded favorable outcomes in the late post-operative phase (greater than 8 weeks). Inconclusive results were yielded for aquatic exercises and for bed exercises without progressive increase and without external resistance (Di Monaco & Castiglioni, 2013).

The second systematic review analyzed 15 articles comparing functional outcomes between center-based and home-based programs. The article concluded significant improvements in strength and function were found using progressive resistance training (PRT) if the intervention was used early, less than 1 month post-surgery in the center-based program and when carried out late, greater than 1 month post-surgery in the home-based program (Okoro et al., 2012).

The third systematic review determined that improvements in gait speed, cadence, and hip abductor strength were noted with physiotherapy compared to the control group. Physiotherapy is defined as restoring the patient's physical health, which encompasses physical therapy and rehabilitation. In addition, the article concluded that no differences were noted in most outcome measures between the outpatient supervised group and the unsupervised home exercise program (HEP) group, except for the Timed Up and Go test, which yielded better results in the supervised group (Coulter et al., 2013).

Lastly, the fourth systematic reviewed analyzed 74 studies and compared deficits in gait mechanics preoperatively to post-operatively in patients who received a total hip replacement for osteoarthritis. The study additionally compared post-operative gait biomechanics to healthy adults. Results determined that patients demonstrated improvements in kinematic gait patterns post-operatively, although deficits were still observed 12 months post-THR when compared to healthy individuals (Bahl et al., 2018). The types of interventions and intervention time were not described in the study. See Table A2 for summary of results.

Randomized Controlled Trials

Two randomized controlled trials were included in this review. One article compared the total hip arthroplasty physiotherapy care protocol (THAPCP) to the total hip arthroplasty protocol (THAP). On the first day, the THAP group received verbal instructions and demonstrations of physiotherapy exercises to be performed in the sitting position once daily for 60 minutes. The exercises focused on strengthening gluteal and thigh muscles. On the second day, the THAP group received instructions and gait training demonstrations. The THAPCP group received the same assistance, plus all physiotherapy exercises and gait-training exercises were performed with the physiotherapy professional. The study determined that the THAPCP group had higher improvements in strength and range of motion compared to the THAP group. The THAPCP group showed improvements in mobility, muscle strength, quality of life, functional capacity, goniometry, and pain at follow-up 15 days post-surgery. This study concluded that the

THAPCP might be a safe approach to accelerate the rehabilitation of THA patients (Marchisio, Galvão, & Borges, 2014).

A second randomized controlled trial compared usual care and usual care plus functional exercise, administered between 12-18 weeks post-operative THA. The functional exercise group performed 12 physiotherapy exercises for approximately 35 minutes, twice a week for 6 weeks. The supervising physiotherapist monitored form and progressed exercise intensity when appropriate. Usual care followed exercises described in the post-operative booklet, which included the following exercises: ankle pumps, static gluteal contractions, static quadriceps, and active hip flexion and abduction and advised patients to ambulate daily with crutches for 6 weeks while progressing distance gradually. At 12 weeks post-operative, no significant differences between the control and intervention group were noted, although at 18 weeks post-operatively, the intervention group demonstrated statistically significant improvements in the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) and walking speed compared to the control group (see Table A3 for summary of results). No significant difference in hip abduction strength were noted (Monaghan et al., 2017).

Non-Randomized Controlled Trials, or Prospective Studies

One case report was included in this review, which assessed the effectiveness of an accelerated four-phase rehabilitation program post hip arthroscopy with a chondroplasty and labral repair. The phases of rehabilitation focused on trunk stabilization through lower extremity kinetic chain exercises, which were based on the regional interdependence model. The third phase focused on restoring muscular and cardiovascular endurance, improving balance and proprioception, and optimizing neuromuscular control. The fourth phase focused on returning to sport. Both the third and the fourth phases of rehabilitation included TRX suspension training exercises, including planks, gluteal bridges, and mountain climbers. After 16 weeks, the high school football player returned to sport pain-free (Cheatham & Kolber, 2012). See Table A4 for summary of results.

A non-randomized pilot study assessed the feasibility of incorporating a closed-chain exercises program into the acute hospital setting. The program was designed for patients who underwent a total knee arthroplasty (TKA) or a total hip arthroplasty (THA), and it focused on functional exercises designed to improve transfers, bed mobility, gait, and stair training. Exercises included were closed kinetic chain knee extensions, mini squats, marching, bilateral plantar flexion, and partial/full step-ups. The article concluded that closed-chain exercises were safe and that they significantly reduced hospital stays to fewer than 4 days (Abbas & Daher, 2017). See Table A5 for summary of results.

Summary of Results

In summary, research shows that early rehabilitation post-operative THR is critical, and data confirm that early intervention significantly improves strength and function. The use of PRT, ergometer cycling, and closed-chain exercises all seem to demonstrate positive effects in the early stages of rehabilitation. At approximately 6 weeks, post-THR surgery improvements were noted in hip ROM, walking speed, step length, and stride length. In the late stages of rehabilitation, many benefits were noted due to exercises performed in weight bearing (greater than 8 weeks post-operatively) and with the use of functional exercises (18 weeks post-operatively), which significantly improved gait speed. Although improvements are noted when compared to preoperative values, patients continue to demonstrate deficits in strength and gait mechanics even 2 years post-surgery when compared to healthy individuals. Refer to Table A1 for Center for Evidence-Based Medicine (CEBM) and Quality Rating of Individual Articles.

Discussion

OA is the main contributing factor leading to THR surgery, due to increased hip pain and a decrease in function contributing to a lower quality of life. Several articles reviewed focus on examining different treatment approaches and comparing results to usual care. Novel approaches appear to be beneficial; however, an enhanced protocol has not been created reflecting the current research.

TRX is a form of suspension training based on functional exercises. Research has suggested that TRX exercises have been beneficial for improving muscular strength, endurance, and cardiovascular health (Smith et al., 2016). In the older population, benefits in balance were noted with TRX exercises, possibly due to increased core activation with use (Gaedtke & Morat, 2015). Further research is necessary to explore the benefits of using suspension training exercises post-THR surgery rehabilitation. Currently, there is no research supporting the benefits of TRX exercises in improving clinical outcomes in the THR population. A limitation in the current literature review is that a single researcher conducted the search and analysis.

Conclusion

An analysis of the articles selected revealed that different rehabilitation approaches seem to be beneficial post-THR. At this time, no specific treatment protocol is more effective than another, and a direct comparison of these results is not possible due to a heterogeneous population and because protocols were conducted during different stages of the healing process. Further research is required to create an enhanced rehabilitation protocol reflecting the current research. In addition, more research is necessary focusing on the benefits of suspension training for the rehabilitation post-THR surgery in the elderly population.

CHAPTER 3

CASE REPORT

Background and Purpose

Hip OA is classified by reactive changes in the bone and degeneration of articular cartilage resulting in pain, stiffness, and overall dysfunction of the joint (Dagenais et al., 2009). In 2010, the prevalence for a total hip replacement was 0.83% of the population in the United States, with an increase in prevalence for the older population. It has been estimated that 5.26% of individuals over the age of 80 have had total hip replacement (THR), which represents approximately 2.5 million individuals in the United States (Kremers et al., 2015). Surgical approaches for a hip replacement include anterior, posterior, and lateral. The posterior approach is the most common approach; it requires an incision of the tensor fasciae latae, iliotibial band, and gluteus maximus muscles with the detachment of the short external rotations. The lateral approach requires splitting the gluteus maximus and vastus lateralis muscles, with elevation of the gluteus medius, minimis, and the anterior portion of the vastus lateralis. The anterior approach has gained popularity in recent years because of the minimally invasive nature of the surgery. This surgical approach involves dissecting between the tensor fasciae latae and sartorious muscles while sparing other muscles (Moretti & Post, 2017).

Hip post-operative precautions vary depending on the type of approach selected. These precautions are enforced to minimize the potential risk for dislocation. In the posterior approach, the contraindications include no hip flexion past 90 degrees, no hip internal rotation, and no hip adduction past midline. With the anterior approach, there is a low chance for dislocation, and, therefore, no precautions are required, although it is suggested to limit passive external rotation and extension. In the lateral approach, the precautions are no hip extension with external rotation actively or passively (Ververeli, Lebby, Tyler, & Fouad, 2009).

The standard THP protocol can be described in four phases. The goal in phase one, 1-5 days post-operative, is to control edema, maintain proper circulation in lower extremities, maintain range of motion, and strengthen hip musculature. The second phase, 5 days to 4 weeks post-operative, includes progression of gait with a cane, stationary cycling, resistance training, closed-chain exercises, balance, and proprioception training. Phase three, 4-10 weeks post-operative, focuses on endurance training, stairs, increasing resistance for closed-chain exercises, and return-to-work tasks. Lastly, the goal for the fourth phase, greater than 10 weeks post-operative, is to return to specific work tasks and recreational sport.

A literature review on THR rehabilitation post-surgery revealed that early interventions are important to decrease hospitals' length of stay and contribute to a quick and successful recovery (Abbas & Daher, 2017). Favorable outcomes have been noted focusing on strength training, ergonomic cycling, and the use of closed-chain exercises in the early stages of rehab. In the later stages (greater than 8 weeks), benefits have been noted with exercises performed in weight bearing. Although many protocols appear to have successful outcomes, deficits in gait mechanics and gait speed have been observed even 2 years post-operative THR (Rasch, Dalén, & Berg, 2010).

Total Body Resistance Exercise (TRX) training is based on closed-chain exercises utilizing body weight and gravity to adjust the resistance. TRX exercises focus on engaging the core and aid in improving strength, flexibility, and overall body stability (Smith et al., 2016). A study investigating the acute and the chronic benefits of TRX suspension training concluded that an 8-week program had positive effects on cardiovascular health, body composition, flexibility, balance, muscular strength, and muscular endurance (Smith et al.). Research has revealed the benefits of using TRX suspension training exercises; however, there is limited evidence on their benefits post-operatively. One case report describes an accelerated post-op labral tear protocol that included TRX suspension training exercises for the rehabilitation of a high school athlete. This patient demonstrated full recovery and returned to sport pain-free 16 weeks after surgery. TRX exercises have been proven to have successful outcomes, although there is limited evidence on the benefits of TRX training in the elderly population and no evidence available on its benefits post-operative THR. The purpose of this case report is to describe the benefits of incorporating TRX exercises in the rehabilitation of an older patient who underwent total hip replacement surgery.

Case Description: Patient History and Systems Review

The patient was a 60-year-old male who presented to physical therapy with complaints of left hip pain. The patient reported that he wanted to be pain-free and return to full work duty without restrictions. The patient had been employed full time at Walmart for approximately 15 years and was recently unable to perform work duties that included prolonged periods of standing, walking, and kneeling to restock produce because of the left hip pain. In June of 2017, the patient was stocking a box weighing approximately 50 pounds when he felt a sudden onset of left hip pain, which prevented him from bearing weight on it. The patient was referred to physical therapy three times a week for 2 weeks, and the physician prescribed anti-inflammatory medication, muscle relaxants, and topical agents. After six visits of therapy, the patient was progressing slower than expected and was referred out for imaging. Conventional radiographs revealed the patient had decreased left femoroacetabular joint space with sclerosis and had developed bone spurs over the femoral head. Findings were consistent with severe osteoarthritis, and no dislocation or fractures were noted. Refer to Figure B1 for radiographic films. Magnetic resonance imaging (MRI) results indicated cyst formations and cartilage loss in the femoral head and the acetabulum. The physician informed the patient of the osteoarthritis and discussed the option of hip replacement. Five months later, in November 2017, the patient underwent left total hip replacement surgery using the posterior approach. Refer to Figure B2 for radiographic films post-surgery. After the surgery, the patient was placed on no-work restrictions and was referred to home health physical therapy. After two sessions of home health physical therapy, the patient was referred back to outpatient physical therapy for post-surgical rehabilitation. See Table B1 for the patient timeline and Figure B3 for the patient International Classification of Functioning, Disability and Health (ICF) model.

Vitals were assessed at the post-operative evaluation; the patient’s blood pressure was 136/80 mmHg with a heart rate of 81 beats per minute. The patient reported that he was taking Lisinopril to manage high blood pressure in addition to cyclobenzaprine, ibuprofen, and acetaminophen. The incision site appeared to be clean, dry, and healing well, although it was tender to palpation. Gait assessment revealed a moderate to severe antalgic Trendelenburg gait pattern and decreased stance phase on left lower extremity with use of a single-point cane. Evaluation yielded that patient demonstrated weakness of gluteus maximus, gluteus medius, and external rotator muscles when compared to contralateral musculature. Straight leg raise (SLR) and hip extension ranges of motion were limited both actively and passively. The patient reported a 2/10 pain level rating on the Visual Analogue Scale (VAS). Balance was noted to be impaired in Sharpened Romberg Test and in Single-Limb Stance Test, with closed-eyes testing, indicating the patient relied heavily on vision for balance. Gait speed was assessed in the 10-Meter Walk Test (10MWT) and the Timed Up and Go test (TUG). Both these tests indicated the patient had diminished gait speed and was at risk for falls. The patient's self-reported Lower Extremity Functional Index (LEFI) score was low, indicating great disability. Refer to Tables B2-B8 for clinical outcome measures taken pre-intervention.

Clinical Impression #1

Three weeks post-total hip replacement surgery, the patient was referred to outpatient physical therapy. Neurological compromise during surgery had been ruled out based upon the patient's subjective report and preservation of the sciatic nerve during surgery. The patient did not experience numbness, tingling, or radiating pain, and pain was centralized over the incision site and to the gluteal region. The objective measures planned for the examination included hip strength, hip range of motion, balance testing, and gait speed. TRX training was determined to be a beneficial tool to aid in the post-operative rehabilitation due to subjective complaints of weakness in hip musculature as well as impairment findings upon examination.

Examination

Range of Motion (ROM)

Hip ROM was measured to assess muscular tightness. Hip extension ROM was measured with the patient positioned in prone with the hip in neutral and knee fully extended. The pelvis was stabilized with one hand to disallow for anterior pelvic tilting. The lower extremity was lifted up off the table, and passive hip extension range of motion was measured. The fulcrum of the goniometer was positioned over the greater trochanter, with the proximal arm over the lateral midline of the pelvis and the distal arm in line with the lateral femur (Norkin & White, 2016). Bilateral hip extension was measured both passively and actively in the same manner. Normal hip extension ROM is 20 degrees, according to the American Association of Orthopedic Surgeons (Norkin & White).

To measure hip abduction, the patient was positioned in supine with bilateral knees extended and hips in neutral. The pelvis was stabilized to prevent pelvic rotation and lateral tilting. The hip was placed into abduction, with the fulcrum over the anterior superior iliac spine (ASIS). The proximal arm was aligned with an imaginary line from ASIS to contralateral ASIS; the distal arm was aligned with the midline over anterior femur using the patella bone as a reference (Norkin & White, 2016). Hip abduction was measured on right and left hip both passively and actively in the same manner. Normal hip abduction ROM is 40 degrees, according to the American Medical Association (Norkin & White).

Hamstring length was measured in supine with bilateral knees in full extension and hips in neutral. The pelvis was stabilized to disallow for lumbar flexion and posterior pelvic tilting. The lower extremity was then lifted until a firm end-feel was determined, while ensuring the knee remained in the full-extended position (Norkin & White, 2016). Active hip flexion was measured in the same manner. Bilateral lower extremities were measured both actively and passively. Normal hamstring length in the straight leg raise (SLR) position is 68.5 degrees in males (Norkin & White).

Hip flexion with the knee bent, hip internal rotation, hip external rotation, and hip adduction ranges of motions were not measured due to contraindications and potential increased risk of prosthesis dislocation. Refer to Table B2 for hip ROM value measures pre-intervention and post-intervention.

Criterion-related validity for goniometric measurements was established when compared to the joint position on gold standard, radiographic image. Studies found that the mean difference between goniometer measurements and radiograph images ranged from 0.5 to 3.3 degrees (Norkin & White, 2016). Measurements of the extremities, using the universal goniometer, have been determined to have good-to-excellent reliability (Norkin & White). The minimum clinically important improvement (MCII) for the hip is 13.3 degrees (Foucher, 2016).

Manual Muscle Testing (MMT)

Manual muscle testing (MMT) was done to determine the strength of hip musculature. Grades ranged from 0 to 5, where 0 = No contraction, 1 = Trace, 2 = Poor (full ROM in a gravity-eliminated position), 3 = Fair (full ROM against gravity), 4 = Good (full ROM against gravity and able to maintain position against moderate resistance), and 5 = Normal (full ROM against gravity and able to maintain position against maximal resistance; Reese, 2013). Refer to Table B3 for MMT values.

The patient was positioned in prone to measure the strength of the gluteus maximus muscle. The posterior aspect of his pelvis was stabilized, and the patient's knee remained in full extension during testing. The patient was taken through full range hip extension passively to determine the available range and to instruct the desired motion. The patient then performed desired motion actively; the gluteus maximus muscle was palpated over the muscle belly during testing prior to applying resistance over distal thigh into hip flexion (Reese, 2013). Bilateral gluteal muscles were tested.

The patient was positioned in prone with bilateral knees extended to test hamstring strength. The posterior aspect of the pelvis was stabilized, and the examiner passively assessed knee flexion ROM. The patient was then instructed to flex knee actively while the examiner palpated the biceps femoris muscle. The patient was asked to maintain knee in 90 degrees of flexion while the examiner applied resistance into knee extension at the posterior aspect of distal lower extremity (Reese, 2013). Bilateral hamstrings were tested during the examination.

The gluteus medius muscle was tested with the patient positioned in side-lying with the uppermost limb being tested. The patient was passively taken into hip abduction and demonstrated the desired position. The gluteus medius muscle was palpated above the greater trochanter, and the lateral aspect was stabilized. Resistance was applied into adduction at distal lateral thigh just proximal to the knee joint (Reese, 2013). MMT was performed bilaterally.

Lastly, the patient's external rotators were tested. The patient was in the seated position off the table with bilateral upper extremities holding onto the table for stabilization. PROM into external rotation determined the patient's available range; the patient performed the desired motion actively while the examiner palpated the piriformis muscle. The examiner applied resistance onto the medial aspect of the distal leg into internal rotation with the contralateral hand over the anterolateral portion of the distal thigh (Reese, 2013). Bilateral hip external rotators were tested.

Studies of the psychometric properties associated with MMT found that interrater and intrarater reliability are adequate to excellent (ICC=0.66-1.00) for the lower extremity muscle in the non-specific patient population (Fan et al., 2010). In patients with osteoarthritis, research found the following values: specificity = 0.90, sensitivity = 0.35, and positive likelihood ratio = 3.5 (Youdas, Madson, & Hollman, 2010). Literature has suggested that the sensitively of MMTs may not be adequate enough to measure strength in good to normal range (Shumway-Cook, Brauer, & Woollacott, 2000). The minimal detectable change is 15N when using a handheld dynamometer and a score of 1 on a 0-5 scale (Mahony, Hunt, Daley, Sims, & Adams, 2009).

Gait Speed

Gait speed was measured with the 10-Meter Walk Test (10MWT) and the Timed Up and Go test (TUG). Refer to Table B4 for gait speed values. In the 10-Meter Walk Test, a 10-meter path was marked at 0, 2, 6, and 10 meters. Two meters before the start line and after to finish line were designed to account for acceleration and deceleration. The patient was given the instructions, "Walk as quickly but as safely as possible" (Bohannon, 1997). Time was recorded when the lead foot crossed the start line and stopped when the lead foot crossed the finish line. The patient ambulated with a single point cane in his right hand. Three trials were performed, and the average time was recorded as gait speed. The preferred score for males in their 60s is 3.11 feet/seconds (Bohannon). The 10MWT has excellent test-retest reliability (r=.75-.90) for comfortable gait speed (Watson, 2002) and interrater reliability (ICC = 0.980; Wolf et al., 1999). 10MWT construct validity has excellent correlation with the 6-minute walk test (correlation coefficient = 0.82) in patients with hip fractures (Wolf et al.). In the geriatric population, a change greater and equal to 0.10 m/s is considered a substantial meaningful change (Perera, Mody, Woodman, & Studenski, 2006).

The TUG test was performed to assess balance, walking ability, mobility, and fall risk. A three-meter path was measured; a chair was placed at one end of the path, and a cone was placed at the opposite end of the path. The patient began the test sitting in the chair and was given the instructions, "Stand up, walk as fast and as safe as you can around the cone, walk back, and have a seat" (O'Sullivan, Schmitz, & Fulk, 2014). Three trials were performed, and the average time was recorded as the gait speed. No assistive device was used in the examination. Normal is considered fewer than 10 seconds; greater than 10 seconds is the cut-off score for hip osteoarthritis that is indicative of fall risk (O'Sullivan et al.). In the elderly population, intertester and intratester reliability are high from .92-.99 (Shumway-Cook et al., 2000). Construct validity (Pearson r=.75) has been determined when correlating TUG scores with gait speed. The minimally detectable change for the TUG is 3.6 seconds (Shumway-Cook et al.).

Balance

Two balance tests were performed, Sharpened Romberg and Single-Limb Stance (SLS). Refer to Tables B5 and B6 for balance values. The purpose of the Sharpened Romberg test was to assess the patient's static balance. The patient removed shoes and socks, and the following instructions were given: "Place one foot directly in front of the other so that your heel touches your toe," and "Cross your arms in front of your chest and maintain this position" (El-Kashlan, Shepard, Asher, Smith-Wheelock, & Telian, 1998). The test was performed with the left foot placed in front of the right and with the right foot placed in front of the left and repeated with eyes closed. Time was stopped if the patient opened his eyes, moved arms, or moved his feet on the floor. Normal balance time is 30 seconds, without sway or unsteadiness (El-Kashlan et al.). Intrarater reliability with eyes opened and closed ICC= 0.99. The MDC for the Sharpened Romberg with eyes open is 9 to 10 seconds, and 3 to 9 seconds with eyes closed (Weightman, 2014).

Single-limb stance was assessed to measure balance on one limb as well as to indicate fall risk. The patient removed shoes and socks prior to testing. The following instructions were given: "Cross your arms over your chest. When I say go, balance on one leg and maintain the position as long as you can" (Mak & Pang, 2009). Testing was stopped if the patient's arms moved, foot touched the ground, or feet moved on the floor (Mak et al.). The test was performed on both right and left leg and repeated with eyes closed. Normal values for a patient between the ages of 60 and 69 years of age are 22.5+/-8.6 seconds with eyes open and 10.2+/-8.6 with eyes closed (Mak et al.). Literature suggests that the SLS test has poor absolute reliability and is not sensitive enough to detect changes in elderly adults. A change of 24.1 seconds is required to be considered a significant change (Goldberg, Casby, & Wasielewski, 2011).

Lower Extremity Functional Index (LEFI)

The Lower Extremity Function Index questionnaire (LEFI) was administered to the patient pre- and post-intervention to determine lower extremity function. The patient was asked to circle the level of difficulty he would have with different functional activates. Grading was assessed using the following scale: extreme difficulty or unable to perform activity was scored as 0, quite a bit of difficulty was scored as 1, moderate difficulty was scored as 2, a little bit of difficulty was scored as 3, and no difficulty was scored 4. All of the scores were added up; scores range from 0 to 80, with lower scores indicating greater disability (Binkley, Stratford, Lott, Riddle, & & North American Orthopaedic Rehabilitation Research Network, 1999). The LEFI has excellent internal reliability (α=0.96) and is a valid tool when compared to the short-form-36 questionnaire. The minimal detectable change and the minimum clinically important difference is 9 points (Binkley et al.). Refer to Table B7 for LEFI scores.

Pain

The Visual Analog Scale (VAS) was used to quantify the amount of perceived pain the patient was experiencing. The patient was asked to rate his pain on the scale ranging from none to extreme amount of pain at the beginning of each session. The VAS is typically 100 mm in length, where the score can be determined by measuring the millimeter, with higher ratings indicating greater amounts of pain. Literature suggests that the test-retest reliability is good (r=0.94, P=0.001). Criterion validity could not be determined, because a gold standard for pain does not exist. A difference of 1.37 cm is considered to be clinically significant (Gould, Kelly, Goldstone, & Gammon, 2001). Refer to Table B8 for pain values.

Clinical Impression #2

Prognosis

The patient demonstrated good prognosis due to being in fairly good health. The patient did have high blood pressure, but it was controlled with medication. The patient reported that, overall, he ate a healthy diet and denied smoking, alcohol, and drug use. He lived in a stable household with good family support, which included his wife and two children. The patient had previous back pain at the time of initial injury, which resolved post-THR surgery; no additional musculoskeletal issues were noted during the initial evaluation. The patient reported he was moderately active and enjoyed playing basketball on the weekends and indicated he was constantly walking at work. Literature suggests that, 90 days post-total hip replacement surgery, the rates of complication were 0.02% for a wound infection, 3.1% for a hip dislocation, and 4.6% for readmission to hospital. Factors that increased risks for complications include increased age, the presence of comorbidities, and low income (Monaghan et al., 2017). Literature additionally suggests that improvements in gait mechanics, gait speed, stride length, hip strength, hip ROM, and function outcomes are noted after total hip replacement when compared to pre-op values (Bahl et al., 2018).

Plan of Care (POC)

The POC for this patient incorporated the use of TRX exercises specifically focused on improving the patient's impairments. The intervention was incorporated into a 4-week program where the patient attended therapy twice a week. POC consisted of a 5-

10 minute warm-up on the elliptical followed by grade I and II posterior to anterior glides of the femoroacetabular joint and myofascial release to iliotibial band, tensor fasciae latae, and gluteus medius muscles. The patient was seen for 1-hour visits, where the majority of the time was spent doing exercises on the TRX. Two hip stretches were incorporated into the POC, a bilateral hip flexor stretch and hip adductor stretch to improve hip ROM. Each stretch was held statically 30 seconds and repeated twice. For strengthening, the patient performed a series of closed-chain exercises with the TRX including mini squats, heel raises, mini lunges, and bridges, three sets of 10 repetitions each. Strengthening exercises were progressed to single-limb squats, squats on a BOSU, isometric squats with hip abduction, and bridges with resisted hip abduction. For balance training, the patient used the TRX to perform repetitions of single-limb hip flexion, hip extension, and hip abduction, performing three sets held for 30 seconds each bilaterally. To decrease fall risk and prepare for ambulation on uneven surfaces, the patient performed step ups onto BOSU in the frontal and sagittal planes, three sets of 10 each. Sprints and lunge hops with TRX were incorporated to address gait speed. Initially, these exercises were performed three sets held for 15 seconds and progressed to three sets of 30 seconds. The patient was encouraged by the examiner during exercises to increase speed. TRX exercises were not performed on the floor, because the patient demonstrated difficulty getting on and off the floor and due to the potential risk of hip dislocation. A home exercise program (HEP) was created to reflect TRX exercises. The HEP included prone hip flexor stretch, standing hip adductor stretch, mini wall squats with swiss ball, standing heel raises, mini lunges, bridges, single-limb balance with a chair, and quick stepping. Stretches were held for 30 seconds each repeated twice, strengthening exercises

were performed for three sets of 10, balance exercise was performed for three sets and held for 30 seconds, and quick stepping was performed three sets held for 15 seconds. The HEP was to be performed twice daily. At the 4-week follow-up, the patient's progress was assessed, and the effectiveness of the intervention was evaluated.

Intervention

TRX exercises were selected because studies have suggested that these exercises can have positive effects on muscular strength, endurance, and cardiovascular health (Smith et al., 2016). A study evaluating the feasibility of a TRX suspension-training program in older adults determined that all participants noted strength gains and slight improvements in balance, possibly due to increased core activation during exercise (Gaedtke & Morat, 2015). The patient selected for this case report had a history of high blood pressure and reported deficits in muscular endurance, muscle strength, hip ROM, and balance post-THR surgery. Therefore, TRX suspension training exercises were selected as an appropriate intervention to aid in the rehabilitation process. The goal of the intervention was to improve hip strength, hip ROM, balance, and gait speed and decrease pain and overall disability to return to full work duties.

The patient attended physical therapy in an outpatient setting twice a week for 4 weeks, with each session lasting 1 hour. The TRX training program tailored for the patient began with bilateral hip flexor and hip adductor stretches to improve hip ROM; refer to Table B9 for description and dosage of stretches. Additional hip stretches were not performed due to the post-operative contraindications. A series of closed-chain exercises was selected to strengthen the hip musculature. Exercises selected included mini squats, heel raises, mini lunges, and bridges. Mini squats and mini lunges were selected because THR precautions disallowed hip flexion passed 90 degrees; therefore, exercises were only performed through a partial range (see Table B9 for exercise descriptions and dosage). The patient reported that the TRX exercises were not challenging enough and that he was not experiencing signs of soreness the following day. The exercises were continually progressed the third week to meet the appropriate challenge level; this level was determined based on the exercise progression protocol, which defines the appropriate level as soreness that is relieved after a warm up. Progressions included squats on a BOSU, single-limb squats, isometric squats with hip abduction, and bridges with resisted hip abduction. Resistance was applied with an orange thera-band tied around the patient's distal thighs. The patient reported that lighter resistance bands were not challenging enough, and the thicker bands were too difficult to perform the prescribed sets and repetitions; however, the patient reported that the orange thera-band was the appropriate level of challenge. See Table B9 for description of exercise progression with dosage. Refer to Table B10 for the list of exercises performed each session.

The patient demonstrated diminished balance, especially with eyes closed during the examination. Single-limb TRX exercises were designed to improve balance. The TRX straps provided some support, although the patient was encouraged to use the least amount of upper extremity support as possible. In single-limb stance, the following exercises were performed: hip flexion, hip extension, and hip abduction. These exercises were selected to improve balance while addressing muscular deficits. Hip extension was selected to strengthen gluteus maximus muscles, and hip abduction focused on isolating gluteus medius muscles. Hip flexion was incorporated into the program to mimic gait, left stance phase in coordination with right limb advancement. Initially, exercises were performed with eyes opened and then progressed to eyes closed. Step-ups in the frontal and sagittal planes were incorporated to mimic ambulation over uneven surfaces and decrease potential fall risk. BOSU is a dome-shaped piece of exercise equipment developed to enhance balance, core strength, and proprioception (Ruiz & Richardson, 2005). Refer to Table B10 for the list of exercises performed each session, and refer to Table B9 for exercise description and dosage.

The patient demonstrated decreased gait speed compared to healthy individuals, increasing his risk of falls. To address this impairment, TRX sprints and lunge hops were added to the program. Refer to Table B9 for description of exercises and for the prescribed dosage. These exercises were selected because they required reciprocal lower extremity motions, which is a component of gait. The examiner encouraged the patient throughout the activity, yelling, “Faster, faster, faster,” “Pretend you’re in a marathon,” and “Keep going!” A HEP was created to reflect the TRX program; see Table B11 for details regarding the HEP.

Additional interventions included a 5- to 10-minute warm-up on the elliptical and grade I and II Posterior to Anterior mobilization of the femoroacetabular joint for pain modulation and to increase blood circulation to the hip joint. Myofascial release with The Stick was performed to his left lateral gluteal muscles based on the patient’s complaints of muscle tightness. The Stick is a tool made of flexible plastic that provides compression to the muscles to treat trigger points and muscle pain (Mikesky, Bahamonde, Stanton, Alvey, & Fitton, 2002). The patient was not working at the time of the study and refrained from physical activities, which included prolonged periods of walking, playing basketball, and going to the gym.

Outcomes

Overall improvements were noted in hip ROM, hip strength, balance, gait speed, and self-assessed lower extremity function. The MCID for the hip is 13.3 degrees (Foucher, 2016); therefore, the patient demonstrated clinically significant improvements in SLR and extension actively and passively as well as active abduction range of motion. No change was noted in passive hip abduction range of motion. See Table B2 for full ROM outcome measures.

Muscle strength improvements were noted in gluteus maximus and gluteus medius muscles, although no improvements were noted in external rotators and hamstring strength; see Table B3 for full outcome measures. Literature suggests that MMTs are clinically useful, although more testing is required to determine their application and validity (Cuthbert & Goodheart, 2007). However, literature has determined that a change of one out of five on an MMT rating scale is a clinically significant change (Mahony et al., 2009).

Gait speed significantly improved in the 10MWT and the TUG test. In the 10MWT, the average time between three trials improved from 5.98 seconds to 5.12 seconds, an overall decrease of .86 seconds or 0.17m/s. Literature suggests that a change greater than or equal to .10 m/s is considered a substantial meaningful change (Perera et al., 2006). In the TUG test, the patient improved from an average of 11.96 seconds to 7.55 seconds, an overall decreased of 4.41 seconds. The minimally detectable change for the TUG is 3.6 seconds (Shumway-Cook et al., 2000). See Table B4 for full details regarding gait speed values.

Improvements in balance were noted in both the Sharpened Romberg Test and in the Single-Limb Stance Test. In the Sharpened Romberg Test, the patient demonstrated the most difficulty balancing with his eyes closed regardless of which foot was positioned anterior to the other. The patient was only able to balance a maximum of 4 seconds with his left foot in front of his right and 2 seconds with his right foot in front of his left when asked to close his eyes. After the 4-week program, the patient was able to maintain position for a maximum of 18 seconds with left foot positioned anterior and 13 seconds with his right foot positioned anteriorly with eyes closed. Refer to Table B5 for Sharpened Romberg results. Improvements were also noted in the single limb bilaterally with eyes open and closed, although greater improvements were noted with eyes open from 1.5 seconds on left to 29 seconds and from 7 seconds on right to greater than 30 seconds. Minimal improvements were noted with eyes closed from 1.5 seconds on left to 3 seconds and 3 seconds on right to 9 seconds. A minimal detectable change of 24.1 seconds is considered a significant change (Goldberg et al., 2011). Therefore, the patient made significant balance improvements in the single-limb stance test with eyes open but not with eyes closed. See Table B6 for full single-limb stance balance outcome measures.

No improvements in pain were noted using the VAS, and the patient actually demonstrated increased pain post-4-week TRX program by a score of 2 at evaluation to a score of 3 post-treatment. See Table B8 for daily VAS scores. The patient complied with the HEP and completed exercises five times a week during the 4-week program with no noted increase in pain or adverse effects. A difference in score of 1.37 is considered to be clinically significant (Gould et al., 2001); therefore an increase from a 2 to a 3 is not a clinically significant change.

Scores on the LEFI increased from 20/80 to 29/80 at the end of the TRX rehabilitation program. A minimum change of 9 points is required to be considered a minimal detectable change (Binkley et al., 1999); therefore, the patient had significant improvements in lower extremity function. See Table B7 for LEFI values.

Discussion

The purpose of this study was to evaluate a 4-week TRX program and compare outcomes to the total hip replacement protocol. The results of this study suggest that TRX-based exercises are an effective tool to aid in the post-operative rehabilitation of a patient who underwent a THR. Statistically significant improvements were noted in hip ranges of motion, hip strength, gait speed, balance, and lower extremity function. Improvements in the TUG and 10MWT scores suggest the patient's gait speed is within normal limits and is no longer categorized as a fall risk (Bohannon, 1997). No statistically significant changes were noted in pain when using the VAS, and the patient reported an increase in pain by one point possibly due to the patient experiencing muscle soreness from the TRX exercises.

One limitation of this study is that it is difficult to quantify how much upper extremity support the patient was using with the straps compared to the usage of the lower extremities. In addition, the angle of the straps can alter the difficulty of the exercise; in this study, the angle was not measured while the patient performed the exercises, because there is no current research available on the correlation between the strap angle and muscle force.

During the second week of the program, the patient was sick and missed a physical therapy appointment. It is not clear whether this instance affected the overall results of the program; however, the patient was compliant for the remainder of the program and with his HEP. THR precautions needed to be obeyed throughout the 4-week program. Because of these precautions in effect during this study, the LEFI score may not accurately reflect the amount of improvement the patient truly made during the program. The score is expected to drastically improve once the precautions are discontinued.

Another limitation of this study is that solely a single subject was assessed during the 4-week TRX rehabilitation program. Additionally, this program was limited to 4 weeks, and a longer protocol may better reflect its benefits. Limited research on the benefits of incorporating TRX exercises in a post-operative protocol is available, and there is currently no evidence suggesting its benefits in patients who underwent a THR.

Based on literature and results of this case study on the THR rehabilitation, it is apparent that an appropriate rehabilitation program is important to decrease disability and return to full function. TRX-based exercises may be an appropriate tool based on the benefits noted in this case report; however, further research is required to adequately determine their effectiveness and to define the most optimal TRX protocol.

CHAPTER 4

DISCUSSION

A literature review revealed individuals who underwent THR surgery demonstrated deficits in hip muscular strength prior to surgery. These deficits improved post-surgery; however, patients continued to demonstrate deficits when compared to healthy individuals (Bahl et al., 2018). A systematic review on THR protocols revealed that ergonomic cycling and maximal strength training exercises were beneficial in the early stages of rehab, while positive outcomes in the late stages were found with weight-bearing exercises (Di Monaco & Castiglioni, 2013). Closed-chain exercises in the acute setting were beneficial for improving bed mobility, transfers, gait, and stair training, overall decreasing patients' length of stay in the hospital (Abbas & Daher, 2017).
TRX is a form of functional training based on closed-chain exercises that has been proven to improve strength, core activation, and balance (Gaedtke & Morat, 2015). Straps are suspended from a single anchor point, creating an unstable environment and further engaging core musculature (Byrne et al., 2014). Studies incorporating the use of TRX exercises for rehabilitation have yielded positive results (Cheatham & Kolber, 2012), though, currently, there are no studies evaluating the benefits of TRX training in the post-operative THR population. The purpose of this case report was to evaluate a patient who was post-operative THR and to compare clinical outcomes to usual care.

Results from the case report suggest significant improvements were noted with the 4-week TRX-based rehabilitation program. The patient significantly increased hip strength in gluteus maximus and gluteus medius muscles. No significant increase was noted in hamstring and hip external rotator strength. The patient made significant improvements in the single-limb stance balance test with eyes open, but not with eyes closed. Additionally, LEFI scores increased by 9 points. These results indicate lower extremity function significantly improved from 25% to 36.3%. Gait speed significantly improved in the TUG and the 10MWT. No significant change in reported pain was noted using the VAS.

A limitation of the literature review is that a single reviewer performed the literature review; therefore, there may have been some bias when selecting studies. Additionally, a single reviewer also assigned each article a quality of appraisal score, which may have decreased the accuracy of the grading. The literature review may not have included all relevant studies. Selecting broader search terms such as adult, total hip surgery, and post-surgical rehabilitation may have included other relevant studies into the review. In addition, there is no evidence suggesting the benefits of TRX in the THR population.

The TRX program yielded comparable results to usual care utilizing the THR protocol. Patients who were 6 weeks to 3 months post-operative demonstrated statistically significant improvements in hip abduction strength, although no improvements were noted in hip extensor strength in the usual care category (Coulter et al., 2013). The patient in this case report also demonstrated statistically significant improvements in hip abductor strength as well as hip extensor strength after the 4-week TRX rehabilitation program.

Hip range of motion (ROM) improved both with the THR protocol and the TRX intervention. Six weeks post-operatively, patients improved hip abduction range of motion from 13.63-13.76 degrees (Bahl et al., 2018) to 15 degrees when utilizing the TRX rehabilitation program. However, the patient in this case report was 9 weeks post-operation, and therefore should demonstrate slightly greater improvements in ROM. Hip extension range of motion only improved by 6.8 degrees 12 months post-operatively with usual care (Okoro et al., 2012) and 15 degrees with the TRX rehabilitation program. The TRX rehabilitation protocol demonstrated greater and quicker improvements in hip extension ROM when compared to usual care.

Pain was quantified utilizing the Visual Analogue Scale (VAS). Literature suggests patients who are 18 weeks post-operative had a mean score that decreased by 1.5 (Monaghan et al., 2017). In this case report, the patient actually increased by a score of 1, from a 2/10 to a 3/10. A change in score of 1 is not a clinically significant change, and the patient reported feeling sore from the exercises. It is not clear at this time whether the TRX rehabilitation program is as effective as usual care in decreasing pain levels.

The TRX rehabilitation program was more effective than the THR protocol in improving gait speed. Eight weeks post-operation, the mean score in the TUG for the patient utilizing usual care was 11.1 seconds; at 12 weeks post-operation, no significant improvements were noted (Okoro et al., 2012). The patient in this case report was 9 weeks post-operation and demonstrated a TUG score of 7.55 seconds, indicating the TRX rehabilitation program was more effective in improving gait speed than usual care.

The positive outcomes that occurred with the TRX rehabilitation program were believed to be due to the program specifically addressing the patient's impairments. The greatest improvement when compared to usual care was gait speed. The THR protocol has not been proven to be effective in increasing gait speed, whereas the results from this case report suggest that TRX exercises are effective. TRX sprints and lunge hops are functional exercises that mimic gait. The patient was encouraged to perform these exercises at a fast pace, and the results suggest that speed was carried over into gait.

A limitation of the case report is the duration of the intervention. TRX suspension training exercises were only performed for 4 weeks and were incorporated only from week six to week nine post-operatively. The benefits of TRX suspension training may have been better assessed if the intervention were performed for a longer period and if the intervention was initiated earlier in the rehabilitation process. TRX training involves upper extremity support, and a limitation is that there is no means of measuring and quantifying the amount of upper extremity support. In addition, the angle of the straps can alter the difficulty and intensity of the exercise, and, in the study, the angles were not taken into account. Additional interventions were used in the study, such as grade I and grade II posterior to anterior femoroacetabular mobilizations and soft tissue mobilizations using "The Stick." During week two of the intervention, the patient missed a physical therapy appointment. It is not clear if missing an appointment influenced any results; however, the patient was compliant with his HEP. Results from this case report cannot be generalized to all patients who have had a total hip replacement. The study was conducted on solely one individual and is not an appropriate representation of the entire THR population.

Future studies are required to validate the effectiveness of TRX in the post-operative THR population. Studies should include a longer follow-up period, a larger sample size, and a control group for direct comparison. In addition, future research is required to establish an appropriate protocol detailing specific exercises used and noting session times and initiation time. Additional interventions should not be utilized during future studies to determine the true effectiveness of TRX intervention.

In summary, using TRX suspension training exercises was effective in improving hip strength, single-limb balance, gait speed, and LEFI score in a 60-year-old male patient who underwent THR surgery.

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APPENDIX A

FIGURES AND TABLES FOR LITERATURE REVIEW

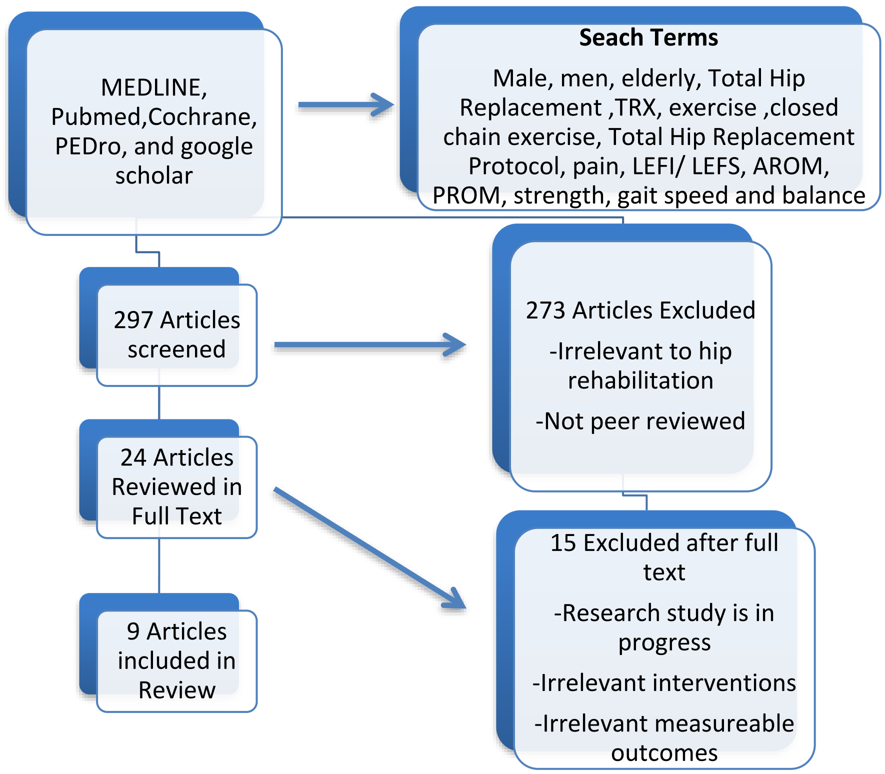

Figure A1. Search strategy.

Table A1

CEBM and Quality Ratings of Individual Articles

Study	CEBM Level of Evidence	Quality
Di Monaco and Castiglioni (2013)	1	92.5% (PRISMA)
Bahl et al. (2018)	1	88.9% (PRISMA)
Coulter, Scarvell, Neeman, and Smith (2013)	1	77.8% (PRISMA)
Monaghan et al. (2017)	2	8/10 (PEDro)
Marchisio, Galvão, and Borges (2014)	2	7/10 (PEDro)
Cheatham and Kolber (2012)	4	26/29 (CARE)
Abbas and Daher (2017)	4	17/22 (STROBE)
Rasch, Dalén, and Berg (2010)	4	17/22 (STROBE)

Note. CEBM indicates Center for Evidence-Based Medicine. PEDro indicates Physiotherapy Evidence Database. CARE indicates case reports. STROBE indicates Strengthening the Reporting of Observational studies in Epidemiology.

Table A2

Systematic Reviews

Authors	Sample Characteristics	Search Strategies	Inclusion Criteria	Results
Di Monaco and Castiglioni (2013)	973 citations were screened, 28 papers assessed, 11 articles included reporting on 9 RCT.	MEDLINE, PEDro, Cochrane Library and Cinahl	The literature was searched since January 2008 until December 2012. Articles included were RCTs; language was English; interventions described the type of exercise and the timing and interventions began after THA. Measurable outcomes included impairments, activity, participation, health-related quality of life or length of stay in the hospital.	- Favorable outcomes for the early post-operative phase were due to maximal strength training and ergometer cycling. - Aquatic exercises led to inconclusive results as well as bed exercises without progressive increase or without external resistance. - Favorable outcomes in the late post-operative phase (>8 weeks) were due to WB exercises.
Okoro, Lemmey, Maddison, and Andrew (2012)	15 RCTs were included. 11 were center-based, two were home-based, and two compared center- and home-based interventions.	MEDLINE, EMBASE, and CINAHL.	Articles included were RCTs that addressed rehabilitation interventions to improve functional outcomes in the THA post-operative population. All publications were in English. Early interventions occurred <1 month post-operatively, and late interventions occurred >1 month post-operatively.	- Significant improvements in strength and function were noted with PRT when carried out early in center-based studies, r late in home-based settings. No significant differences noted between center- and home-based programs.

Table A2, continued

Authors	Sample Characteristics	Search Strategies	Inclusion Criteria	Results
Coulter, Scarvell, Neeman, and Smith (2013)	3,096 articles screened, 32 read in full text and 5 studies comprised of 234 participants were included.	MEDLINE, CINAHL, EMBASE, PEDro, and the Cochrane Library	- Design: RCTs and English language -Participants: Adults after a THR. - Interventions: Physiotherapist-directed rehabilitation exercises. - Outcomes measured: Gait, function, muscle strength, and quality of life.	Physiotherapy rehabilitation improved gait speed by 6m/min, cadence by 20 steps/min, and hip abductor strength with a mean of 16 Nm. Function and quality of life were inconclusive due to a heterogeneous population.
Bahl et al. (2018)	74 studies with a total of 2,477 patients were included.	PubMed, MEDLINE, CINAHL, The Cochrane Library, Embase, Scopus, Web of science, Sport Discus and Health collection.	Adults over the age of 18 undergoing a THA with OA being the primary indication. Studies reporting on changes in gait from pre-operative to post-operative, who used 2D and 3D motion analysis techniques.	6 weeks post-operative improvements were noted in walking speed, stride length, step length, and hip ROM compared to preoperative gait. Hip ROM increased in all planes at 3 months. 12 months post-operatively patients demonstrated deficits for walking speed, stride length, single-limb support, and sagittal hip ROM compared to healthy individuals.

Note. Abbreviations: WB, weight bearing; RCT, randomized control trial; THA, total hip arthroplasty; PRT, progressive resistance training; THR, total hip replacement.

Table A3

Randomized Controlled Trials

Authors	Participants	Intervention	Control	Outcomes Measured	Results
Marchisio, Galvão, and Borges (2014)	106 patients with hip OA who underwent THA surgery.	N=54 THAPCP group received the same verbal instructions and demonstrations guided by a physiotherapist.	N=52 The THAP group received verbal instructions and physiotherapy exercise demonstrations.	- Hip ROM using goniometry -Hip strength -The discrepancy of the lower limbs -Quality of life: Medical Outcomes Study 36-Item Short Form Health Survey (SF-36) -Motor, gait and pain performances: Merle d'Aubigné and Postel scores	THAPCP demonstrated greater improvements in ROM and strength compared to the THAP group. THAP group improved functional capacity, quality of life, mobility, muscle strength goniometry, and pain. The THAPCP group had a significant improvement compared to the THAP group ($p=0.007$ vs. $p = 0.10$) in the global clinical evaluation.

Table A3, continued

Authors	Participants	Intervention	Control	Outcomes Measured	Results
Monaghan et al. (2017)	63 subjects ≥50 years who underwent a THR for osteoarthritis- Inclusion criteria: Understand English, willing to attend class two times a week for 6 weeks, and able to perform exercise program without assistance. -Exclusion criteria: Suspicion of infection, medically unstable, or other terminal illness.	N=32 Usual care plus functional exercise, administered between 12-18 weeks post op THA. The program included 35-minute sessions 2x per week for 6 weeks. The physiotherapist whom corrected form and progressed exercises taught 12 exercises.	N= 31 Usual care: patients were given a post-operative exercise book to follow for 6 weeks. Book included LE exercises as well as recommended walking progression.	WOMAC questionnaire, VAS, 6MWT, SF 12, and isometric hip abduction strength test. Outcomes were measured 12 and 18 weeks post-operatively.	-Week 12: No significant difference between the control and intervention groups. -Week 18: Intervention group demonstrated statistically significant improvements in WOMAC (P<0.01), and walking speed (P<0.04) compared to the control group. There was no difference in hip abductor strength.

Note. Abbreviations: OA, osteoarthritis; THA, total hip arthroplasty; THAPCP, total hip arthroplasty physiotherapy care protocol; THAP, total hip arthroplasty protocol; ROM, range of motion; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index questionnaire; VAS, visual analogue pain scale score; SF 12, Short Form, self-reported physical and mental health scores; 6MWT, 6-minute walk test.

Table A4

Case Report

Authors	Participants	Intervention	Outcomes Measured	Results
Cheatham and Kolber (2012)	18-year-old high school football player who underwent left hip arthroscopic surgery for mixed FAI with an anterior superior labral tear.	Four-phase rehabilitation program. - Phase III: Included exercises using the TRX. Exercises incorporated were single-limb squats, side lunges, suspended planks, and gluteal bridges. - Phase IV: Included mountain climbers using a TRX.	Hip pain, ROM, strength, muscle tenderness using the five-point pain scale, muscle length using Ober's test, and gait assessment.	The patient returned to playing football after 16 weeks and was pain-free with no signs of FAI at 4-month follow-up.

Note. Abbreviations: FAI, Femoroacetabular impingement; TRX, total body resistance exercise; ROM, range of motion.

Table A5

Other Studies

Authors	Participants	Intervention	Outcomes Measured	Results
Abbas and Daher (2017)	N=535 patients who underwent a THR from 2011-2013.	NPORP: closed-chain exercises tailored to focus on meeting discharge goals (independent bed mobility, transfers, gait, and stair mobility). Exercises included knee extension, hip and knee flexion combined with end range knee extension, bilateral PF	Hospital length of stay	Using the NPORP protocol patient's hospital length of stay reduced to 4 days with a significant P-value <0.05.

Table A5, continued

Authors	Participants	Intervention	Outcomes Measured	Results
Rasch, Dalén, and Berg (2010)	20 elderly patients with unilateral OA who underwent a THA. Patients were excluded if they had a prior LE surgery, neurological involvement, or cardiopulmonary disease.	Observation, no intervention.	Objective measures were taken prior to having the THA, and 6 and 24 months after surgery. Objective measures included voluntary isometric strength of LE muscles, gait analysis, postural stability and clinical scores (HHS, EuroQoL, and SF-36)	- Preoperatively, all LE muscles showed a deficit in strength (9-27%, $p < 0.03$), except for knee flexors. 6 months post-operatively, deficits were 8-16% ($p < 0.03$) except for knee flexors and hip adductors. 2 years post-operatively, all muscles showed significant improvements, although hip abductors showed a 15% deficit ($p < 0.001$) - Preoperatively, gait analysis identified a decreased stance time in the OA limb ($p < 0.001$). After 6 months and 2 years, no significant differences noted between limbs. - HHS, EQ-5D, and VAS showed improvements at 2 years follow up.

Note. Abbreviations: THR, total hip replacement; (NPORP), new post-operative rehabilitation pathway; PF, plantar flexion; OA, osteoarthritis; LE, lower extremity; THA, total hip arthroplasty; HHS, Harris Hip Score; SF-36, Short form-36 questionnaire; EQ-5D, EuroQol- 5 Dimension; EuroQoL, EuroQol- 5 Dimension; VAS, visual analogue scale

APPENDIX B

FIGURES AND TABLES FOR CASE REPORT

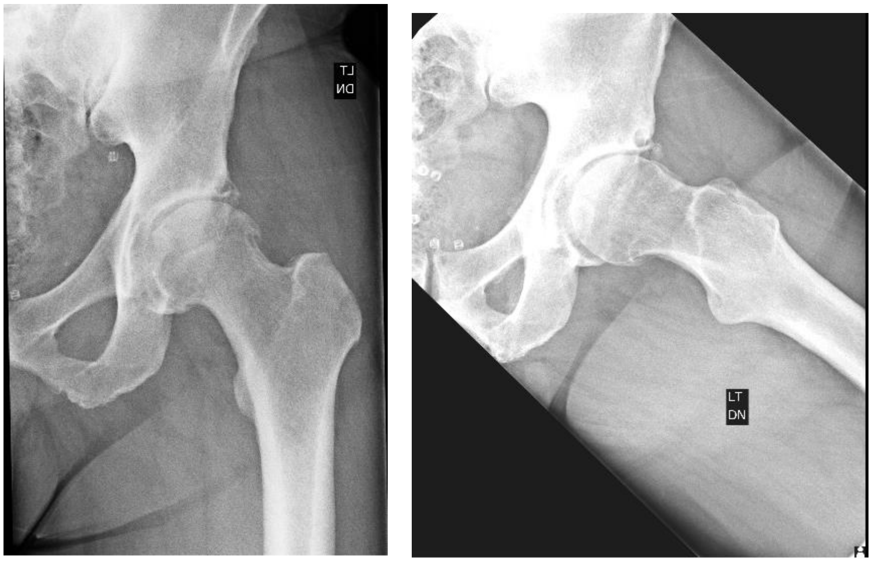

Figure B1. Radiographs prior to THR surgery.

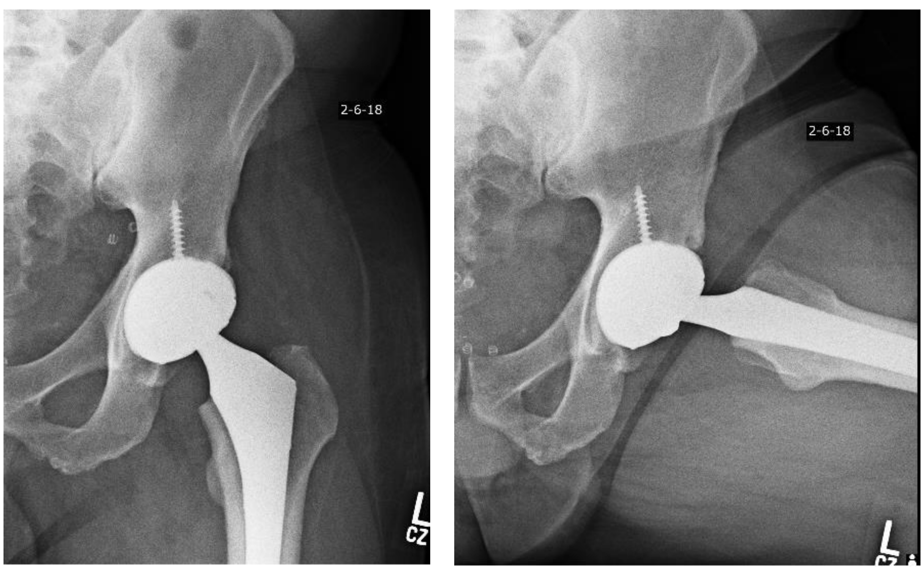

Figure B2. Radiographs post-THR surgery.

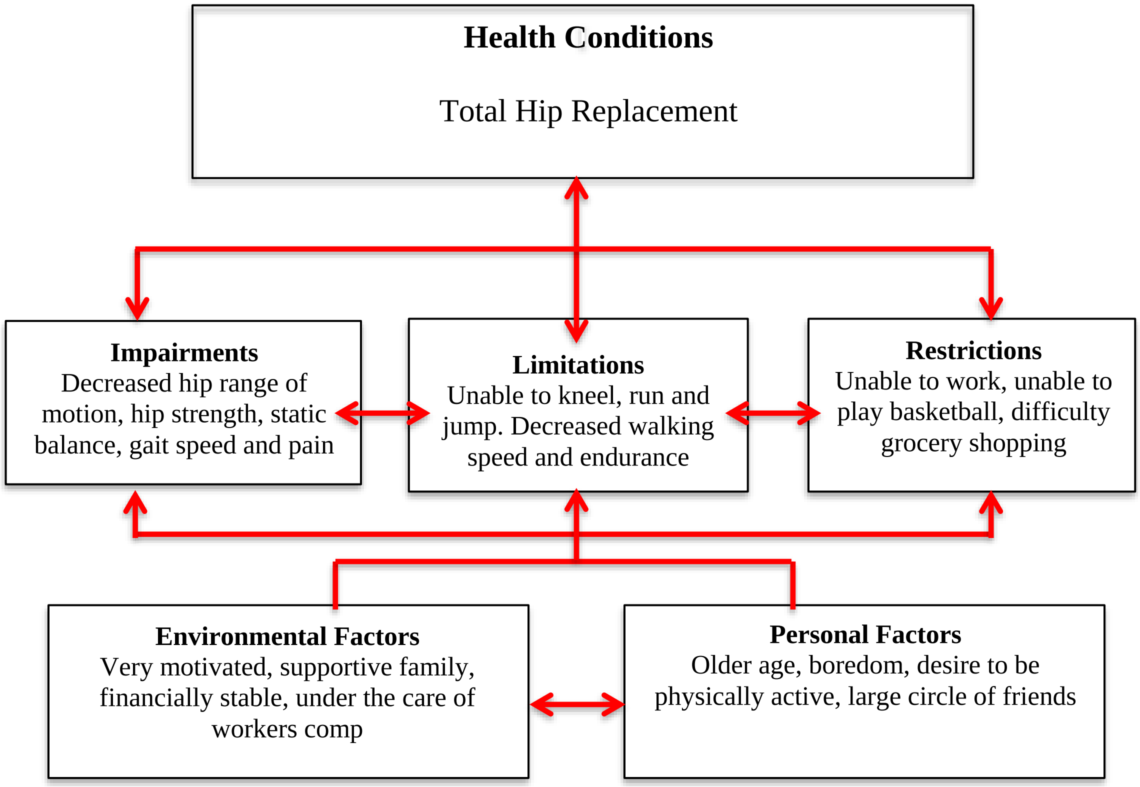

Figure B3. International classification of functioning, disability and health (ICF) model.

Figure B4. Intervention.

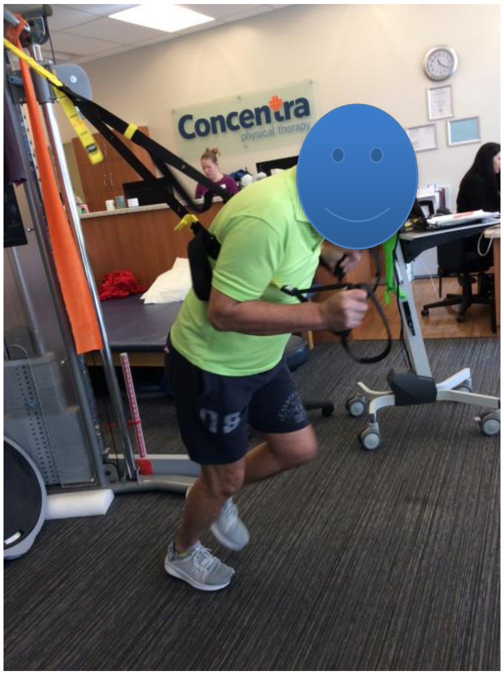

Figure B5. Intervention.

Table B1

Timeline

6.5 months prior	1.5 months prior	1 month prior	Day 1
Date of initial injury and referred to physical therapy.	Date of left total hip replacement surgery and referred to home health physical therapy post-surgery.	Referred to outpatient physical therapy; initial evaluation and treatment.	Incorporation on TRX suspension training.

Table B2

Range of Motion (ROM)

	Pre-test (degrees)	Post-test (degrees)
SLR AROM	L: 20 R: 45	L: 45 R: 55
SLR PROM	L: 25 R: 50	L: 45 R: 55
Hip extension AROM	L: 5 R: 15	L: 25 R: 22
Hip extension PROM	L: 10 R: 20	L: 25 R: 22
Hip abduction AROM	L: 10 R: 15	L: 25 R: 30
Hip abduction PROM	L: 20 R: 20	L: 25 R: 30

Note. Abbreviations: SLR, straight leg raise; ROM, range of motion; AROM, active range of motion; PROM, passive range of motion.

Table B3

Manual Muscle Testing (MMT)

Muscle	Pre-test grade	Post-test grade
Hamstring	L: 4+/5 R: 4+/5	L: 4+/5 R: 5/5
Gluteus maximus	L: 3-/5 R: 3+/5	L: 4/5 R: 4/5
Gluteus medius	L: 3+/5 R: 4+	L: 4+/5 R: 5/5
Hip external rotators	L: 3/5 R: 4/5	L: 3/5 R: 5/5

Table B4

Gait Speed

Test	Pre-test time (seconds)	Post-test time (seconds)
10MWT with SPC	Trial 1: 6.08 Trial 2: 6.11 Trial 3: 5.74 Average time: 5.98	Trial 1: 5 Trial 2: 5.25 Trial 3: 5.11 Average time: 5.12
TUG without AD	Trial 1: 11.80 Trial 2: 12.45 Trial 3: 11.63 Average time: 11.96	Trial 1: 7.53 Trial 2: 7.93 Trial 3: 7.18 Average time: 7.55

Note. Abbreviations: 10MWT, 10-Meter Walk Test; TUG, Timed Up and Go test; SPC, single point cane; AD, assistive device.

Table B5

Sharpened Romberg

Condition	Pre-test time (seconds)	Post-test (seconds)
Eyes open	L foot anterior to R: 30 R foot anterior to L: 30	L foot anterior to R: >30 R foot anterior to L: >30
Eyes closed	L foot anterior to R: 4 sec R foot anterior to L: 2 sec	L foot anterior to R: 18 sec R foot anterior to L: 13 sec

Note. Abbreviations: L, left; R, right.

Table B6

Single-Limb Stance

Condition	Pre-test time (seconds)	Pre-test time (seconds)
Eyes open	L: 1.5 R: 7	L: 29 R: >30 sec
Eyes closed	L: 1.5 R: 3	L: 3 R: 9

Note. The best of three trials reported.

Table B7

Lower Extremity Functional Index (LEFI)

Test time	Score	Implication
Pre-test score	20/80	25% function
Post-test score	29/80	36. 3% function

Table B8

Visual Analog Scale (VAS)

Treatment Session	VAS score
1st	2/10
2nd	2/10
3rd	3/10
4th	2/10
5th	2/10
6th	3/10
7th	3/10

Table B9

TRX Exercise Dosage and Description

Stretches	Repetitions and sets	Description
Hip flexor	2x30 seconds bilaterally	The patient is standing in a staggered stance holding straps with bilateral UEs and facing the anchor. The patient was instructed to shift weight forward to feel stretch in the anterior hip.
Hip adductor	2x30 seconds bilaterally	The patient is standing in a wide stance holding straps with bilateral UEs and facing the anchor. The patient was instructed to bend one leg while shifting weight over that side to feel a stretch on the medial aspect of the thigh.

Strengthening exercises	Repetitions and sets	Description
Mini squats	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor. The patient was instructed to flex bilateral knees and hips to perform a partial squat and reminded not to flex hip past 90 degrees.
Heel raises	3x10	Patient in standing holding straps with bilateral UEs and facing away from the anchor with body aligned at a 45-degree angle. The patient was asked to raise heels slowly off the ground and then back to the ground.
Mini lunges	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor. The patient was standing with one foot in front of the other to perform a lunge and reminded not to flex hip past 90 degrees.
Bridges	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor with trunk at a 45-degree angle from the ground. The patient was asked to flex and extend his hips while contracting his gluteus muscles.
Single-limb squats	3x10 each	Patient in standing holding straps with bilateral UEs and facing the anchor. Patient lifted one leg off the ground while flexing contralateral knee and hip into a squatting position. The patient was reminded not to flex hip past 90 degrees.

Table B9, continued

Strengthening exercises	Repetitions and sets	Description
Squats on a BOSU	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor. The patient was instructed to flex bilateral knees and hips to perform a partial squat and reminded not to flex hip past 90 degrees while standing on a BOSU.
Isometric squats with hip abduction	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor. The patient was instructed to flex bilateral knees and hips to perform a partial squat and instructed to isometrically hold position while stepping laterally five steps to the right and five steps to the left. The patient was reminded not to flex hip past 90 degrees.
Bridges with resisted hip abduction	3x10	Patient in standing holding straps with bilateral UEs and facing the anchor with trunk at a 45-degree angle from the ground. The patient was asked to flex hips and extend hip while abducting against the resistance of an OTB. Thera-band ties around distal thighs.

Balance exercises	Repetitions and sets	Description
Single-limb hip flexion	3x30 seconds bilaterally	Patient in standing holding straps with bilateral UEs and facing away from the anchor. The patient was instructed to balance on one limb while repeatedly flexing the contralateral hip.
Single-limb hip extension	3x30 seconds bilaterally	Patient in standing holding straps with bilateral UEs and facing away from the anchor. The patient was instructed to balance on one limb while activating gluteal muscles and extending the contralateral hip.
Single-limb hip abduction	3x30 seconds bilaterally	Patient in standing holding straps with bilateral UEs and facing away from the anchor. The patient was instructed to balance on one limb while repeatedly abducting contralateral hip.
Step up onto BOSU	3x10 each	Patient in standing holding straps with bilateral UEs and facing the anchor. The patient was instructed to step up onto BOSU with one LE at a time in the frontal plane and then step back to original position. The patient was positioned laterally to BOSU and instructed to step onto BOSU in the frontal plane with one LE at a time and step off onto the contralateral side of BOSU.

Table B9, continued

Gait speed exercises	Repetitions and sets	Description
Sprints	Initially, 3x15 seconds and progressed to 3x30 seconds in the third week	Patient in standing holding straps with bilateral UEs and facing away from the anchor with trunk angled 45 degrees from the ground. The patient was asked to alternate lifting knee up toward his waist at a fast pace.
Lunge Hops	Initially, 3x15 seconds and progressed to 3x30 seconds in the third week	Patient in standing holding straps with bilateral UEs and facing away from the anchor with trunk angled 45 degrees from the ground. Patient standing in a staggered stance in the mini-lunge position and alternating LEs by hopping.

Note. Abbreviations: UEs, upper extremities; BOSU, Bottom Side Up; HEP, Home exercise program; OTB, orange thera-band; LE, Lower Extremity.

Table B10

Exercise Prescription

6-7 weeks post-operative	6-7 weeks post-operative
Stretches	
Hip flexor	Hip flexor
Hip adductor	Hip adductor
Strengthening	
Mini squats	Single-limb mini squats
Mini lunges	Mini lunges
Heel raises	Heel raises
Bridges	Bridges with resisted hip abduction
	Mini squats on a BOSU
	Isometric squats with hip abduction
Balance	
Single-limb hip flexion	Single-limb hip flexion with eyes closed
Single-limb hip extension	Single-limb hip extension with eyes closed
Single-limb hip abduction	Single-limb hip abduction with eyes closed
	Step ups on BOSU in the frontal and sagittal planes
Gait speed	
Sprints	Sprints
Lunge Hops	Lunge Hops

Table B11

HEP Description

Exercises	Repetitions and sets	Description
Prone hip flexor and quadriceps stretches	2x30 seconds bilaterally	Patient in prone with the strap looped around the foot. The patient was asked to bend his knee and pull strap until a stretch is felt over the anterior thigh.
Standing hip adductor stretch	2x30 seconds bilaterally	The patient is standing in a wide stance and instructed to bend one leg while shifting weight over that side to feel a stretch on the medial aspect of the thigh.
Mini wall squats with swiss ball	3x10	Patient in standing with a swiss ball positioned between back and wall. The patient was instructed to bend hips and knees into a mini squat.
Standing heel raises	3x10	Patient in standing holding onto a chair for stability; the patient was asked to raise heels slowly off the ground and then back to the ground.
Mini lunges	3x10	The patient was standing with one foot in front of the other to perform a lunge and reminded not to flex hip past 90 degrees.
Bridges	3x10	Patient positioned in supine with bilateral knees flexed. Patient instructed to lift bottom off the mat while contracting gluteal and abdominal muscles.
Single-limb balance	3x30 seconds	Patient in standing holding onto a chair for stability; the patient was asked to balance on one limb with the least amount of upper extremity support as possible.
Quick stepping	3x15 seconds and 3x30 seconds in the third week	Patient in standing holding onto a chair for stability. The patient alternated lifting knee up toward waist at a fast pace.

Table B12

Clinical Reasoning for Rehabilitation following a Total Hip Replacement

Period	Short-Term Goals	Clinical Challenge/decision	Modification to the rehabilitation program
6 to 9 weeks post-surgery	Increase hip AROM and PROM	Post-operative THR precautions apply	No hip flexion >90 degrees, no hip adduction, no hip internal rotation Limited to grade I and II mobilizations for pain modulations to prevent dislocation of the prosthesis
	Improve hip strength	Severe hip muscular weakness, Trendelenburg gait pattern	Initiation of gluteal strengthening exercises in weight bearing. Resistance band applied over distal thighs for gluteus medius activation and strengthening
	Improve balance	Decreased stance time on the affected limb during ambulation	Initiation of balance exercises, specifically focusing on single-limb balance and progressed with eyes closed Balance challenged with use of BOSU to mimic uneven surfaces
	Increase gait speed	Decreased gait speed in the 10MWT and TUG test	Timed TRX sprints and lunge hops incorporated into POC with encouragement from the examiner.

Note. Abbreviations: THR, Total Hip Replacement; BOSU, Bottom Side Up; 10MWT, 10-Meter Walk Test; TUG, Timed Up and Go test; TRX, Total Body Resistance Training; POC, Plan of Care